

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [70, 69, 75, 68, 66, 75, 70, 75, 81, 83]

Tap amplitudes (pixels): [207.19314660480447, 187.44865963777923, 170.53152201279386, 173.3666634621547, 170.4494059831245, 164.83021567661677, 176.58142597679972, 182.87700784953805, 167.70509831248424, 176.7766952966369, 174.6682569902156]

Gemini 3 Pro

Based on the analysis of the extracted features:

1. **Amplitude:** The subject starts with a very large tap (207 pixels) and immediately drops to a lower range (187, 170). However, the amplitude does not continue to decrement progressively; it stabilizes around 170-180 pixels and even recovers to 182 pixels near the end. This pattern suggests an initial adjustment rather than the pathological "sequence effect" (progressive decrementing) described in scores 2 or 3.
2. **Rhythm/Speed:** The tap periods show some irregularity (fluctuating between 66 and 75 frames) and a distinct slowing trend towards the end of the sequence (increasing to 81 and 83 frames).
3. **Interruptions:** There are no significant outliers in the periods that would indicate a freeze or a complete halt.

The presence of **slight slowing** towards the end and minor rhythm irregularities fits the criteria for a score of 1. The amplitude drop, while starting early, does not persist as a progressive decline, and the movements remain large.

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